

EAM 310 · EAST ASIAN MEDICINE DIAGNOSIS I · VUIM SUMMER 2026

Pre-Lecture Analysis Sheet

Week 3 · Pulse Diagnosis 脈診
FCM Ch.25 · CAM Ch.10 · 28 Pulse Qualities

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Quiz 3 · July 31, 2026

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溫故知新 · 온고지신

01 Behavioral Objectives · What You Will Be Able to Do 目标

BEHAVIORAL ANALYSIS FRAMEWORK

These objectives follow the DO-CONDITION-CRITERION format. Each one names a measurable behavior — not "understand," but discriminate, identify, explain, sequence, predict. Practice each behavior, not just the content.

OBJECTIVE 1 — CLASSIFY

Given any of the 28 pulse qualities by name, classify it into its correct grouping (Depth / Rate / Strength / Size / Length / Shape / Rhythm).

Criterion: Correct on ≥26 of 28 without reference.

OBJECTIVE 2 — DISCRIMINATE

Discriminate between easily-confused pulse pairs: Floating vs. Hollow; Wiry vs. Tight; Fine vs. Minute; Knotted vs. Hasty vs. Intermittent; Slippery vs. Choppy.

Criterion: State feel AND indication for both members of each pair.

OBJECTIVE 3 — SEQUENCE

Sequence the 6-Step Pulse Reading Protocol in correct order from palpation to position-specific findings.

Criterion: Reproduce all 6 steps without prompting.

OBJECTIVE 4 — PREDICT

Given a clinical presentation, predict the expected pulse quality and confirm with organ position (e.g., "Liver qi stagnation → Wiry at Left Guan").

Criterion: Correct quality + correct position.

OBJECTIVE 5 — NAME (SEASONAL)

Name the expected normal pulse quality for each of the four seasons, including approximate date ranges per Schrier's slides.

Criterion: All 4 correct without notes.

OBJECTIVE 6 — IDENTIFY (CLASSICAL)

Identify which Six-Climatic-Qi pathogen produces which pulse signature (Wind→Floating, Cold→Tight+Deep, etc.).

Criterion: Correct for all 6 climatic factors.

BA REINFORCEMENT SCHEDULES FOR THIS WEEK

Continuous reinforcement on new qualities (first exposure: check answer every time). → After 3 correct-in-a-row, shift to **fixed-ratio-3** (check every 3rd). → By day of class, use **variable interval** spacing (random-order flashcard review). This matches acquisition → fluency → retention.

02 Slide Deck Content Survey · Wk_3_EAM_Dx_1__Pulses.pdf (18 slides) 講義

Slides 1–3 · Introduction to Pulse Diagnosis

Overview — The Four Pillars & Pulse Slides 1–2

- ▶ **Pulse diagnosis (脈診 Mài Zhěn)** = one of the Four Pillars (Observation, Listening/Smelling, Questioning, Palpation)
- ▶ Palpation is the **FOURTH** pillar — pulse is its primary tool
- ▶ Schrier's framing: "The deepest, most rewarding skill in EAM"
- ▶ Purpose: assess Qi, Blood, Yin, Yang, and identify organ patterns through the radial artery

Normal Pulse Defined Slide 3

- ▶ **Stomach-Qi (胃氣):** gentle, calm, slightly slow quality — reflects Middle Jiao health
- ▶ **Spirit (神):** soft but with strength, regular rhythm — reflects Heart/Mind
- ▶ **Root (根):** clearly felt at deep level at Rear (Chi) position — reflects Kidney essence
- ▶ Exam pearl: A pulse can show pathological qualities yet still carry Spirit and Root → reasonable prognosis

Slides 4–6 · Positions, Levels, and Method

Cun-Guan-Chi Positions (Left and Right) Slide 4

- ▶ **Left Cun (寸):** Heart / Pericardium
- ▶ **Left Guan (關):** Liver / Gallbladder
- ▶ **Left Chi (尺):** Kidney-Yin / Bladder
- ▶ **Right Cun (寸):** Lung
- ▶ **Right Guan (關):** Spleen / Stomach
- ▶ **Right Chi (尺):** Kidney-Yang / Large Intestine
- ▶ Finger assignment: Index=Cun, Middle=Guan, Ring=Chi

Three Depth Levels & Method for Taking Pulse Slides 5–6

- ▶ **Superficial (浮):** light contact — Wei Qi level
- ▶ **Middle (中):** moderate pressure — Ying Qi level
- ▶ **Deep (沉):** heavy pressure near bone — Yuan Qi / organ level
- ▶ Wait 1 minute minimum; patient rests 5–10 min before examination
- ▶ Patient seated; arm at heart level, palm up, slight dorsiflexion on wrist pillow
- ▶ Schrier: practice on yourself between classes — 3 levels, both hands, 6 positions

Slides 7–14 · The 28 Pulse Qualities — Core Content

EXAM FOCUS — SCHRIER'S SLIDE EMPHASIS

The 28 qualities are grouped into **six classification schemes** in FCM. Schrier's deck emphasizes: (1) the Depth group, (2) Rate qualities, (3) Rhythm irregulars (Knotted/Hasty/Intermittent), and (4) clinically common pairs. He marked quality-vs-quality discrimination as high-yield.

#	Quality / 脉	Feel / Sensation	Key Indication(s)	Group
1	Floating 浮 Fú	Found with light pressure; disappears with heavy	Exterior patterns; also Yin def (Empty at depth)	Depth
2	Deep 沉 Chén	Requires heavy pressure near bone	Interior; +Weak=Qi/Yang def; +Full=stasis/Cold/Heat	Depth
3	Hidden 伏 Fú	Deeper than Deep; extreme pressure, almost no pulse	Severe interior Cold, extreme pain, complete collapse of Yang	Depth
4	Firm 牢 Láo	Deep, Full, Wiry, and Long simultaneously	Interior Cold with Excess — masses, stagnation deep in interior	Depth
5	Leather 革 Gé	Hard on surface, Empty inside (like a drum)	Deficiency of Essence/Blood + internal Cold	Depth
6	Slow 遲 Chí	≤3 beats/breath (~60 BPM)	Cold; +Weak=Empty-Cold; +Full=Full-Cold	Rate
7	Rapid 數 Shuò	≥5 beats/breath (~90+ BPM)	Heat; +Full=Full-Heat; +Empty-Float=Empty-Heat (Yin def)	Rate
8	Slowed-down 緩 Huǎn	4 beats/breath; relaxed, leisurely	Normal (if with Stomach-Qi); Dampness if pathological	Rate
9	Hurried 疾 Jí	≥7 beats/breath; extremely fast	Extreme Heat; exhaustion of Yin with Yang about to separate	Rate
10	Empty 虛 Xū	Big but soft, no strength; collapses under pressure	Qi deficiency (and often Blood)	Strength
11	Full 實 Shí	Full, hard, long, pushes back at all levels	Full/Excess pattern — pathogenic factor present	Strength
12	Weak 弱 Ruò	Deep, Soft, and Fine simultaneously	Deficiency of both Qi and Blood	Strength
13	Scattered 散 Sǎn	Irregular, diffuse, like floating leaves dispersing	Collapse of Yuan Qi; very serious — terminal pattern	Strength
14	Overflowing 洪 Hóng	Surges up under finger like a wave; forceful and wide	Extreme Heat; Summer = normal Overflowing	Size
15	Fine 細 Xì	Thin as a thread; always palpable	Blood deficiency; Dampness with severe Qi def	Size
16	Minute 微 Wēi	Extremely faint; even thinner than Fine	Severe deficiency of BOTH Qi and Blood; critical	Size
17	Long 長 Cháng	Extends beyond normal position boundaries	Heat; Liver yang rising; Full pattern	Length
18	Short	Does not fill the normal position; truncated	Severe Qi deficiency; also Qi stagnation	Length

#	Quality / 脉	Feel / Sensation	Key Indication(s)	Group
	短 Duǎn			
19	Slippery 滑 Huá	Smooth, rolling, oily — “pearls rolling on a plate”	Phlegm, Dampness, food retention, pregnancy (normal)	Shape
20	Choppy 澀 Sè	Rough, jagged, hesitant — “knife scraping bamboo”	Blood deficiency, fluid exhaustion, Blood stasis	Shape
21	Wiry 弦 Xián	Taut like a guitar string; doesn’t wave	Liver disharmony, pain, Phlegm	Shape
22	Tight 緊 Jǐn	Taut AND twisted — like a rope being wrung	Cold, pain; exterior = Wind-Cold; interior = Cold stagnation	Shape
23	Hollow 芤 Kōu	Felt superficially + deeply but EMPTY in middle (scallion)	Hemorrhage or sudden massive blood loss	Shape
24	Soggy 濡 Rú	Floating, Fine, and Soft simultaneously	Dampness; also Blood/Yin def in old/weak patients	Shape
25	Moving 動 Dòng	Round like a bean, short, trembles, no head/tail	Shock, fright, anxiety, extreme pain	Rhythm
26	Knotted 結 Jié	SLOW + irregular stops	Cold obstructing Heart, Heart-Yang def, Yin excess	Rhythm
27	Hasty 促 Cù	RAPID + irregular stops	Extreme Heat; Heart-Qi def; Heart-Fire	Rhythm
28	Intermittent 代 Dài	Stops at REGULAR intervals, any rate	Serious Zang organ dysfunction — most serious of the three	Rhythm

CRITICAL EXAM TRAP — RHYTHM PULSES

Knotted = SLOW + irregular. Hasty = RAPID + irregular. Intermittent = ANY rate + REGULAR stops. Schrier has tested “Knotted is rapid” as a false stem before — do NOT flip the rate.

03 Critical Discrimination Practice · Easily-Confused Pairs 辨别

Depth Pair: Floating vs. Hollow

Feature	FLOATING 浮	VS	HOLLOW 芤
Feel at surface	Felt easily with light pressure	≠	Also felt at surface
Feel at middle	Present (may be weaker)	≠	EMPTY — no pulse at middle
Feel at deep	Disappears or weakens	≠	Present again at deep
Analogy	—		Like pressing a scallion stalk (hollow center)
Indicates	Exterior; Yin def (if Empty at depth)	≠	Hemorrhage ; sudden blood loss

Shape Pair: Wiry vs. Tight

Feature	WIRY 弦	VS	TIGHT 緊
Sensation	Taut like a bowstring — tense but straight	≠	Taut AND twisted like a tightly wrung rope
Movement	Doesn't wave or undulate	≠	Feels like it could snap; more forceful
Primary indications	Liver disharmony, pain, Phlegm	≠	Cold (exterior = Wind-Cold; interior = Cold stagnation), pain
Seasonal note	NORMAL in Spring	≠	Never a normal seasonal pulse

Size Pair: Fine vs. Minute

Feature	FINE 細	VS	MINUTE 微
Strength	Thin but always palpable; has some force	≠	Extremely faint; barely detectable
Severity	Blood def or Dampness with Qi def	≠	Severe deficiency of BOTH Qi AND Blood; critical state
Clinical implication	Commonly seen; not immediately serious	≠	Critical — near exhaustion of vital substances

Rhythm Triple: Knotted vs. Hasty vs. Intermittent

Feature	KNOTTED 結	HASTY 促	INTERMITTENT 代
Rate	SLOW	RAPID	Any rate
Stops	IRREGULAR	IRREGULAR	REGULAR intervals
Indicates	Cold; Heart-Yang def; Yin excess	Extreme Heat; Heart-Fire; Heart-Qi def	Serious Zang organ failure; most severe
Severity	Moderate	Moderate-severe	Most serious

MNEMONIC — RHYTHM PULSES

"KKNOTTED is KOLD (slow) + irregular · HHASTY is HOT (rapid) + irregular · IINTERMITTENT is INTERVAL (regular stops) = most serious"
The K = Cold = Slow. The H = Heat = Rapid. The I = Interval = Regular = serious.

04 Seasonal Pulses 季節脈 · High-Yield (Schrier Flagged) 季節

DIRECT SCHRIER QUOTE FROM SLIDES

"Good quiz question — wink wink nudge nudge." Seasonal pulses appeared on Quiz 2 matching section and will appear again. Know all 4 cold.

Spring (Feb 4 – May 4)

Pulse: WIRY (弦)

Yang Qi wants to arise but cold Yin obstructs its release
Slightly tense; if too wiry → pathological → check emotional state
Not wiry enough (too relaxed) = also pathological

Summer (May 5 – Aug 6)

Pulse: FLOODING/OVERFLOWING (洪)

Yang Qi uninhibited — spreads upward and outward
Full, surging wave quality is normal this season

Autumn (Aug 7 – Nov 6)

Pulse: HAIR-LIKE / FLOATING (毛)

Light and empty; "dainty and delicate, like elm pods falling"
Yang Qi desiring to be restrained Heat and collected

Winter (Nov 7 – Feb 3)

Pulse: STONE-LIKE / DEEP (石)

Yang Qi gathered and stored — sinks back down
When Yang emerges pulse floats; when it retracts, it sinks

MNEMONIC — SEASONAL PULSES

W-F-H-S · "Wires Float Here in Snow"
Wiry (Spring) · Flooding (Summer) · Hair-like (Autumn) · Stone-like (Winter)
Think: Spring sprouts need wire to grow; Summer floods; Autumn hair blows; Winter sinks like stone.

05 Six Climatic Qi Pulse Signatures · Schrier Slide Content 六氣

Pathogen (六淫)	Pulse Signature	Rationale
Wind (風)	Floating	Exterior invasion — Wei Qi rushes to surface
Cold (寒)	Tight + Deep (contracted)	Cold congeals and contracts — pulls pulse inward and tightens
Summer-Heat (暑)	Floating + Vacuous/Empty	Heat rises and expands but exhausts fluids, leaving pulse empty
Dampness (濕)	Soggy + Slippery	Dampness obstructs Qi movement — soggy/slick sensation
Dryness (燥)	Short + Hasty + Rough/Choppy	Dryness depletes fluids — pulse lacks smooth flow
Fire/Heat (火)	Restless + Rapid	Heat accelerates everything — classic rapid pulse

06

Classical Sources · Seasonal Pulses Slide Deck Content

古典

Shang Han Lun Pulses — Six Stages

Stage (六經)	Expected Pulse
Taiyang 太陽	Floating
Yangming 陽明	Flooding (洪) or Long (長)
Shaoyang 少陽	Wiry (弦)
Taiyin 太陰	Moderate at middle depth — neither fast nor slow
Shaoyin 少陰	Faint and Thin
Jueyin 厥陰	Thin with tendency to expire

SCHRIER NOTE ON LINGSHU CONTENT

Schrier presented Lingshu 4 and Nanjing difficulty 58 content as supplementary classical material. The Lingshu organ-quality tables (Heart, Lung, Liver, Spleen, Kidney × pulse quality) are historically important context. Schrier marked the Nan Jing/Shang Han Lun slide as **BONUS — Will Not Test**. Focus energy on FCM Ch.25 content; classical tables are enrichment.

Nanjing Difficulty 58 — Five Kinds of Cold Harm

五種傷寒 Wǔ Zhǒng Shāng Hán

- ▶ 1. Struck by Wind (中風 Zhōng Fēng)
- ▶ 2. Harmed by Cold (傷寒 Shāng Hán)
- ▶ 3. Damp-Warmth (濕溫 Shī Wēn)
- ▶ 4. Heat Disease (熱病 Rè Bìng)
- ▶ 5. Warmth Disease (溫病 Wēn Bìng)

07

The 6-Step Pulse Reading Sequence · Schrier Protocol

程序

BA RATIONALE — WHY SEQUENCE MATTERS

Reading a pulse without a sequence is like a physical exam without a protocol — you'll miss findings and make false conclusions. The sequence builds a behavior chain: each step becomes the discriminative stimulus for the next. Practice the chain, not just the steps.

Six-Step Protocol (from Schrier + FCM Ch.25)

- ▶ **Step 1 · Overall Impression:** Get a global sense of the pulse before analyzing anything. Fast/slow? Strong/weak?
- ▶ **Step 2 · Stomach-Qi:** Does it have a gentle, calm, slightly slow quality underneath the pathology?
- ▶ **Step 3 · Spirit:** Is there softness with strength? Regular rhythm? Or does it feel mechanical/flat?
- ▶ **Step 4 · Root:** Feel the Rear (Chi) position at deep level — can you feel it clearly? Root = Kidney reserves.
- ▶ **Step 5 · Overall Quality:** Name the predominant quality across all positions (Wiry? Slippery? Rapid?)
- ▶ **Step 6 · Individual Positions:** Systematically assess each of the 6 positions for position-specific findings.

MNEMONIC — 6-STEP SEQUENCE

"Overall Snapshots Surface Really Quick" → Overall · Stomach-Qi · Spirit · Root · Quality · (Individual) Positions
Or more clinically: Big picture first → Health markers → Then pathology → Then position-specific.

DANCE MEDICINE APPLICATION — TCM FOR BALLROOM DANCERS

Pulse patterns common in competitive ballroom dancers:

Dancers during peak training often show a **Wiry or slightly Tight pulse** — especially at the Left Guan (Liver/Gallbladder). This reflects: (1) Liver overwork from repetitive sinew strain; (2) Emotional tension of competition performance; (3) Pain from musculoskeletal overuse.

Post-competition: Watch for **Empty or Fine pulse** indicating Blood and Yin depletion after sustained aerobic exertion. Latin dancers especially: sustained hip work + adrenal activation → may present with Rapid + Empty (Yin def pattern).

Ballroom-specific Hollow pulse flag: Female competitors presenting with amenorrhea (common in lean athletic women) may show Hollow pulse — indicates Blood/Essence depletion. Immediately investigate menstrual history alongside pulse finding.

08 Quiz 3 Prediction · July 31, 2026 測驗

HIGH-YIELD TOPICS BASED ON SCHRIER SLIDE EMPHASIS + COURSE PATTERN

Based on Quiz 1 and Quiz 2 formats: expect 6–7 MC, 4–6 T/F, 1 matching set, and possibly a clinical case (describe the pulse → name the pattern).
Schrier tested seasonal pulses twice — they WILL appear again.

VERY HIGH Seasonal pulses — match season to quality; know date ranges (Feb 4/May 5/Aug 7/Nov 7)

VERY HIGH Knotted vs. Hasty vs. Intermittent — rate + regularity + indication; classic T/F trap

HIGH Normal pulse: 3 qualities (Stomach-Qi, Spirit, Root) — what each reflects

HIGH Six pulse classification schemes — given a quality, name its group

HIGH Wiry vs. Tight — sensation difference + different indications

HIGH Slippery pulse — 4 indications (Phlegm, Dampness, food retention, pregnancy)

MODERATE Floating + Tight = Wind-Cold; Floating + Rapid = Wind-Heat (exterior differential)

MODERATE Deep + Weak vs. Deep + Full — deficiency vs. excess differentiation

MODERATE Six climatic Qi → pulse signatures (especially Wind=Floating, Cold=Tight+Deep)

MODERATE Fine vs. Minute — severity distinction; Hollow indication (hemorrhage)

LOWER Shang Han Lun six-stage pulses — Taiyang=Floating, Shaoyang=Wiry (Schrier presented but flagged bonus content for Nanjing material)



PRE-CLASS BEHAVIORAL CHECKLIST

Before class Thursday, verify you can do each of these **without notes**:

- ☐ Write all 28 quality names with pinyin in 5 min
- ☐ State the feel AND indication for each of the 6 discrimination pairs
- ☐ Recite the 4 seasonal pulses with date ranges
- ☐ Sequence the 6-step reading protocol from memory
- ☐ State which climatic Qi produces which pulse
- ☐ Discriminate Knotted/Hasty/Intermittent on 3 flashcards without error